Top-Down Topological Device Using LiDAR

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Abstract

This paper provides schematics for a 1-dimensional top-down topological graphing device using time-of-flight (TopTopToF) with a LiDAR sensor. The device will output segmented and averaged linear topological data as hues to an LED strip through a Raspberry Pi Pico MCU with MicroPython. There are limitations inherent in the rotating mechanism approach and the LiDAR sensor range resolution; error cases include out-of-bounds objects, adjacent object overcasting, and objects of similar heights (less than 1cm difference) within the same sampling segment.

1 Introduction

This project aims to construct a 1-dimensional topological scanning device using a rotating LiDAR sensor. One a larger scale, this device could feature applications in identifying characteristic topography, search and rescue, and more.

2 Parts and Assembly

Table 1: Parts Required

Function	**Part Name**	**Price**	**Power Draw**
Digital Input	Benewake TF Mini-S LiDAR	44$	140mA $\times$ 5V = 0.7W
Microcontroller Unit	Raspberry Pi Pico	10$	100mA $\times$ 5.5V = 0.55W
Digital Output	BTF WS2812B LED Strip (using 10 LEDs)	14$	(10×60)mA $\times$ $5V$ = 3W
Analog Input	Sparkfun 2-Axis I2C Joystick	7$	10mA $\times$ 3.3V = 0.033W
Actuator	SG90 180-degree Rotation Servo	In Lab	700mA $\times$ 5V = 3.5W
Frame Support	Wood planks	?$	
Total		94$	7.783W $\times$ 1.5 $\approx$ 12W Required

Table 2: 12W exceeds possible supply by Pico via USB from Vbus; an external 5V supply is needed.

3 Assembly

The LiDAR will be rigged to the shaft of a 180-degree rotating servo on a wooden frame support structure. As the joystick is activated in one direction of its x-dimension, the servo shaft and LiDAR will rotate. The LiDAR will scan the surface in real-time, recording point-cloud data. This data will then be sent to the microcontroller via I2C which will then output the data (segmented and averaged in code) as height-corresponding hues to respective LED's on the strip. The current LED the LiDAR is projecting onto will blink to indicate that it is being updated real-time, while the other LEDs will keep their previous hue data.

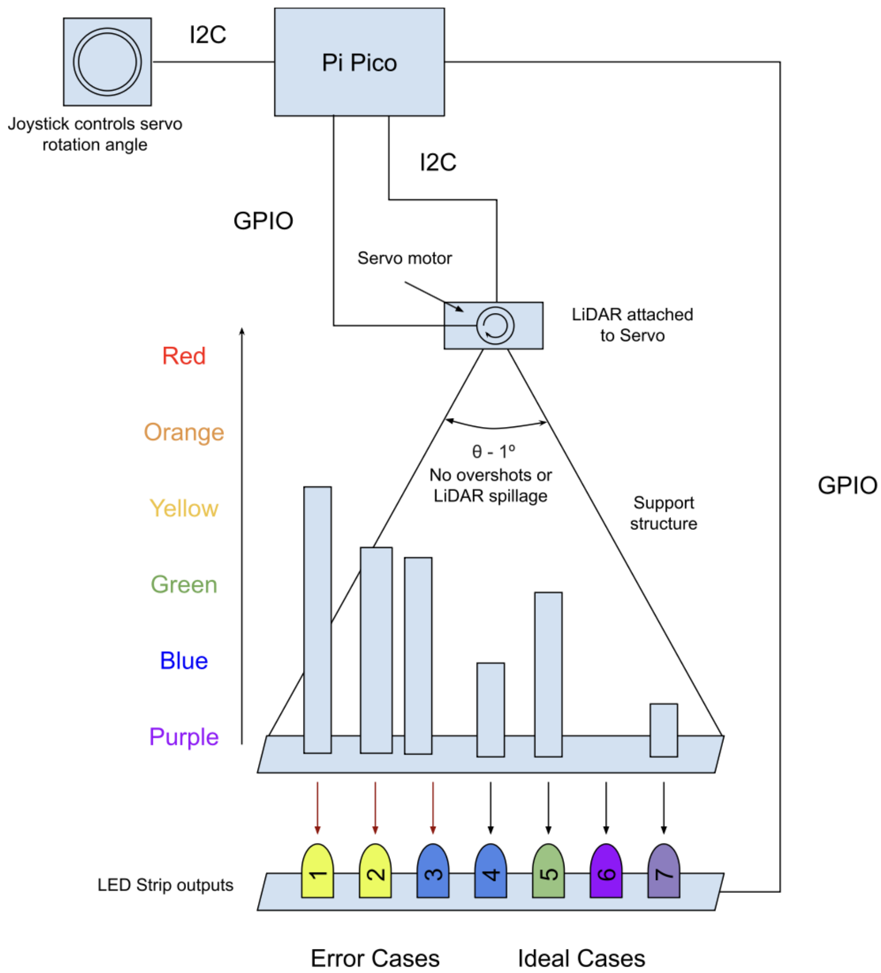

Figure 1: A physical overview schematic of the device.

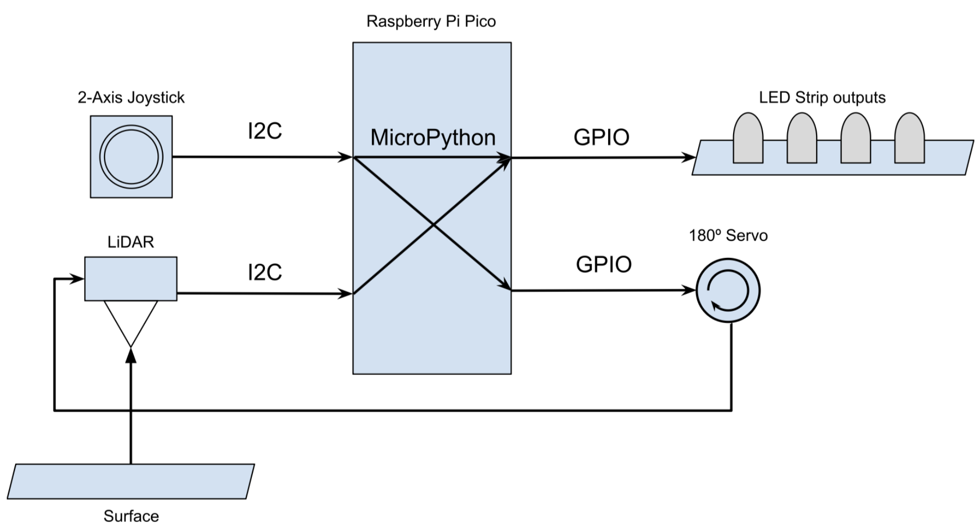

Figure 2: A logical left-to-right overview schematic of the device.

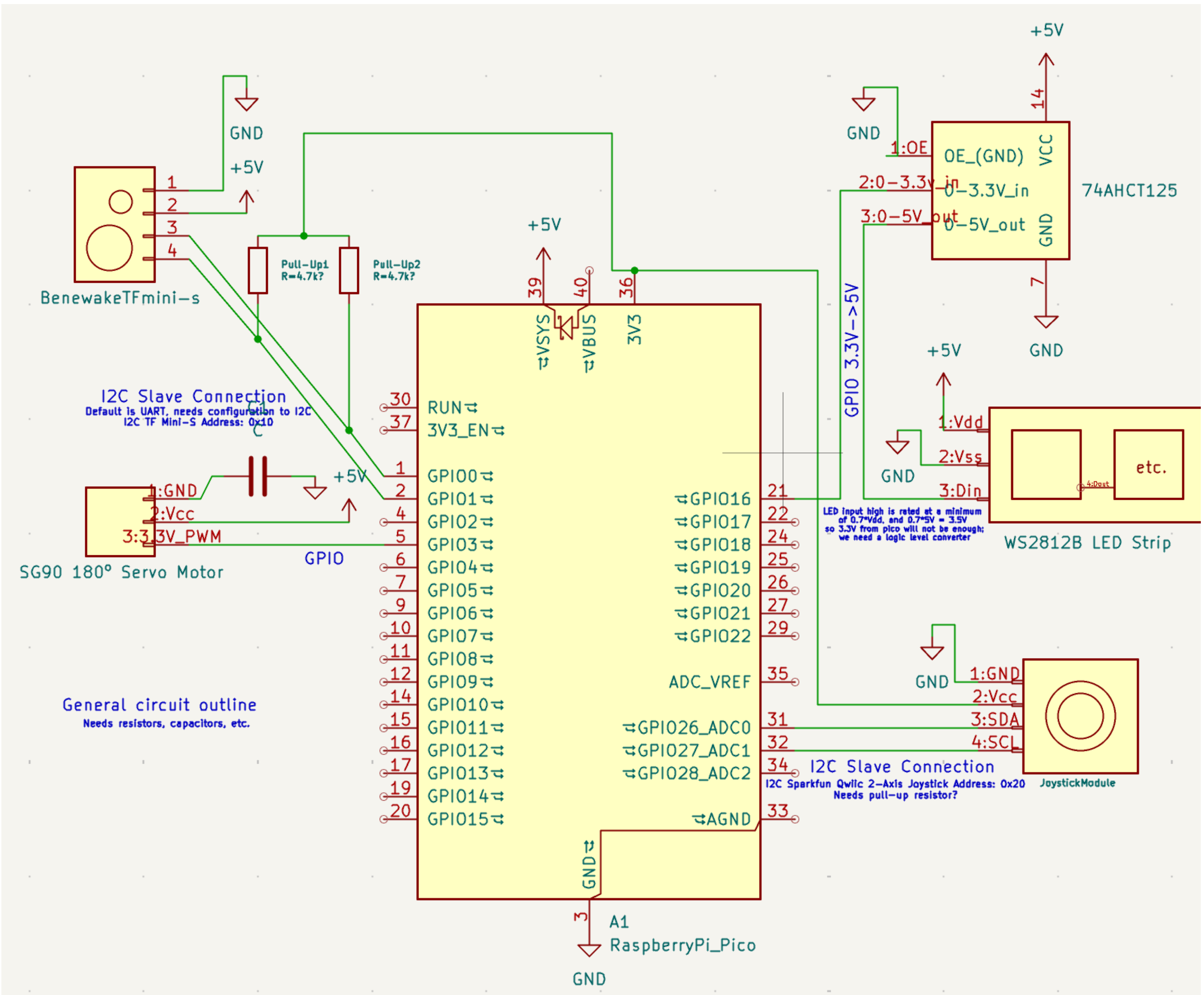

Figure 3: A circuit schematic.

4 Part Specs

4.1 Benewake TF Mini-S

This device is a LiDAR module that detects surfaces within a range of $0.1 \rightarrow 12$ m with an accuracy of $\pm$6cm at ranges from a $0.1 \rightarrow$ 6m resolution. It draws an average current of ≤ 140 mA and power of ≤ 0.7W at 5V.

The TF Mini-S uses the Time-of-Flight (ToF) method for detection and ranging. The device sends an infrared light signal ($\tilde{8}$50nm) and records the amount of time it takes from its emission to its sensing with a photosensor and timed circuit.

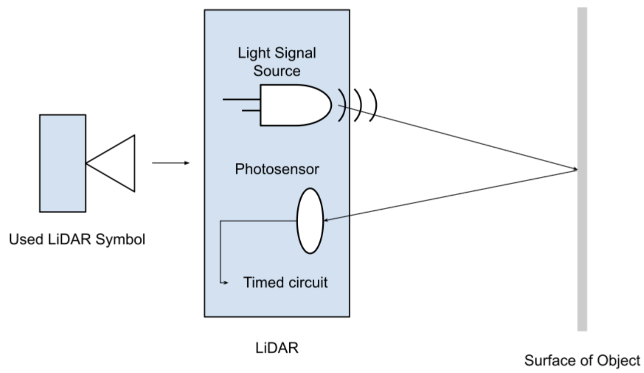

Figure 4: Above is a diagram of the Time-of-Flight LiDAR method; light signals (usually infra-red) are sent my the LiDAR device, reflected by the surface or object, and collected by the LiDAR's photosensor, using the time that it took for the signal to return to measure the distance of the surface or object.

Table 3: TF Mini-S Pinouts

Pin	Name	Function
1	GND	Ground
2	Vcc (+5V)	Supply Voltage
3	RXD/SDA	Receive/Data
4	TXD/SCL	Transmit/Clock

Table 4: This module uses UART by default but is compatible with I2C (must use pull-up resistors); the default I2C address is 0x10.

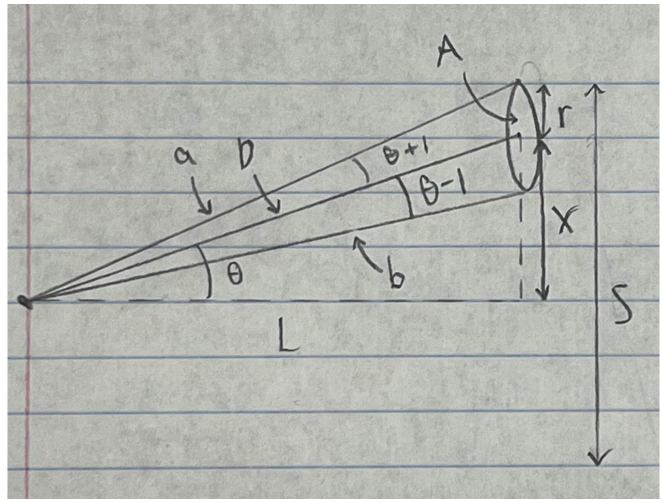

Figure 5: This is the geometry of the LiDAR sensor (left) trajectory to surface (right):

1. L is LiDAR/Servo height
2. D is LiDAR-to-surface distance
3. a is sample area extrema distance
4. b is sample area inner distance
5. x is sample area center to surface center distance
6. r is radius of sample area
7. S is length of surface
8. θ is angle of LiDAR from surface center
9. θ_+ is outer bound angle
10. θ_- is inner bound angle.

Sample Calculation

Let $L = 40$cm, $x_{max} = 30$cm, Strip LED/cm = 1.67, LED/cm Scaling Factor (SF) = 1/10

$$\theta = \arctan(\frac{30}{40}) = 36.87 \tag{1}$$

$$\theta_- = \theta - 1 = 35.87 \rightarrow_{plug} \tan\theta_- = \frac{x_{max} - r}{L} \tag{2}$$

$$\theta_+ = \theta + 1 = 37.87 \rightarrow_{plug} \tan\theta_+ = \frac{x_{max} + r}{L} \tag{3}$$

$$r = 1.08\text{cm} \tag{4}$$

$$A = \pi r^2 = 3.64\text{cm}^2 \tag{5}$$

$$S = 2(x_{max} + r) = 62.15\text{cm} \approx 60\text{cm} \tag{6}$$

$$\#_{\text{LED's}} = S \times \frac{\text{LED}}{\text{length}} = 60\text{cm} \times 1.67\ \frac{\text{LED}}{\text{cm}} = 100 \tag{7}$$

$$\#_{\text{LED's}} \times \text{SF} = 10 \text{ LEDs} \tag{8}$$

A setup of 10 LED's with 6cm surface length per LED.

4.2 Raspberry Pi Pico

For the microcontroller of this project, a Raspberry Pi Pico (RP2350) will be used. This MCU allows for MicroPython–a preferable coding language. The Pico operates with 3.3V logic, natively compatible with the Benewake LiDAR and the Sparkfun Joystick through I2C, but not the WS2812B detailed below.

The on-board Vbus +5V will not be able to sufficiently power all devices, so an external power supply must be used and may be plugged into Vsys.

Table 5: Employed Pi Pico Pinouts

Pin	Name	Function
1	GPIO 0	SDA to TFMini-S
2	GPIO 1	SCL to TFMini-S
21	GPIO 16	SDA to WS2812B LED
31	GPIO 26	SDA to Qwiic Joystick
32	GPIO 27	SDA to Qwiic Joystick
36	3V3	+3.3V logic
39	Vsys	Externally Supplied +5V

4.3 BTF WS2812B LED Strip

This LED strip will output hues dependent on surface-to-sensor distance. Since this strip uses +5V logic, a logic level shifter is required for the Pico (+3.3V) to properly interface with it–a 74HCT125 will work. The MicroPython neopixel library assists with assigning hues to the the LEDs.

Table 6: WS2812B Pinouts

Pin	Name	Function
1	Vcc (+5V)	Supply Voltage
2	GND	Ground
3	DIN	Data In

Table 7: Vcc operates at 4.8-6V, so we will use 5V.

4.4 Sparkfun 2-Axis Analog Joystick

To control the servo and for possible 2-D compatibility in future augments, an analog joystick will be implemented.

Table 8: Sparkfun Joystick Pinouts

Pin	Name	Function
1	GND	Ground
2	Vcc (+3.3V)	Supply Voltage
3	SDA	Data
4	SCL	Clock

Table 9: This module is compatible with I2C; Default I2C address is 0x20.

4.5 SG90 180º servo

For smoothly rotating the LiDAR sensor with precise, knowable rotation angles (for easier LED projection), a 180º servo is preferable.

Table 10: SG90 180º Servo

Pin	Name	Function
1	GND	Ground
2	Vcc (+5V)	Supply Voltage
3	PWM (+3.3V)	Pulse-Width Modulation

Table 11: The pulse-width modulation for controlling this servo may be done with MicroPython's built-in machine.PWM class.

5 5-Week Plan

This progress of this project will be paced with this schedule. Implementation of additional features (2-D upgrades, etc.) would require optimistic progress.

1. Source parts in-house (saw 2-axis joystick, SG90 180º servo) and test with Arduino. Write pseudocode outline for reading LiDAR data input, surface partitioning/averaging, and LED output.

2. Optimistic parts arrival; test components and retest in-house parts with Pico in MicroPython. Begin sourcing materials for frame–will go to Ace Hardware in person and ask around for spare wood.

3. Pessimistic parts arrival; do as in optimistic. Begin the assembly of apparatus–translate circuit schematic to breadboard.

4. Testing and troubleshooting; verify and identify ideal, error, and edge cases.

5. Final testing and troubleshooting. Begin write-up and presentation of the project.

6 Coding and Sources of Info

The Thonny IDE will be used for interfacing with the Pico and may be downloaded here. This video by NerdCave explains how to interface WS2812B addressable LED's using the Raspberry Pi Pico. This video by Circuit Digest shows how to interface the Raspberry Pi Pico W with an Analog Joystick Module. This web tutorial by SunFounder explains how to control a servo with MicroPython.